

Subject 25

The input files are located in the *dataIN* directory.
All the output files will be saved in the *dataOUT* directory.

Requirements

The *Buget.csv* file contains information on the execution of revenue and expenditure budgets at the locality level. The information is as follows:

Siruta - Siruta Code of the locality;

City - Name of the city;

V1, ..., V5 - Revenues

V1 - Income from quotas broken down from income tax

V2 - Amounts allocated from the income tax quotas to balance local budgets

V3 - Amounts broken down from VAT

V4 - Amounts broken down from VAT to balance local budgets

V5 - Subsidies

C1, ..., C8 - Expenses

C1 - Personnel expenses, C2 - Goods and services, C3 - Social assistance, C4 - Public services, C5 - Public order, C6 - Education, C7 - Health, C8 - Culture.

The *LocPopulation.csv* file contains the population by locality and the county codes for each locality.

1. Save in the *Requirement_1.csv* file the incomes and expenses per inhabitant for each category of incomes and expenses. For each locality, the Siruta code, income, and expenses per inhabitant, by category (V1, ..., C1, ...) will be saved. **(1 point)**

2. Save in the *Requirement_2.csv* file the percentages of expenses for each category in total expenses, at the county level. For each county, the county identifier and the percentages for each expenditure category (C1, ...) will be saved. **(2 points)**

The *DataSet_25.csv* file contains records for 97 countries around the world on indicators related to the business environment. The observed variables are as follows:

1. Start-up procedures,
2. Tax payments,
3. Time required,
4. Time to pay taxes,
5. Tax rate,
6. Revenue,
7. Foreign investment,
8. High-tech exports.

3. Determine and save in the *StdCov.csv* file the variance-covariance matrix corresponding to the standardised numerical variables of the *DataSet_25.csv* file. **(1 point)**

4. Conduct principal components analysis on the dataset from *DataSet_25.csv* and compute the scores (standardised principal components). Save the scores into the *Scores.csv* file. **(2 point)**

5. Plot the variance explained by the principal components using a line plot, along with a horizontal line corresponding to the variance equal to one. **(1 points)**

6. Determine the factor loadings matrix and plot the correlogram along with the factor loadings values (2 decimal precision), employing the heatmap plot from the seaborn package. **(2 points)**